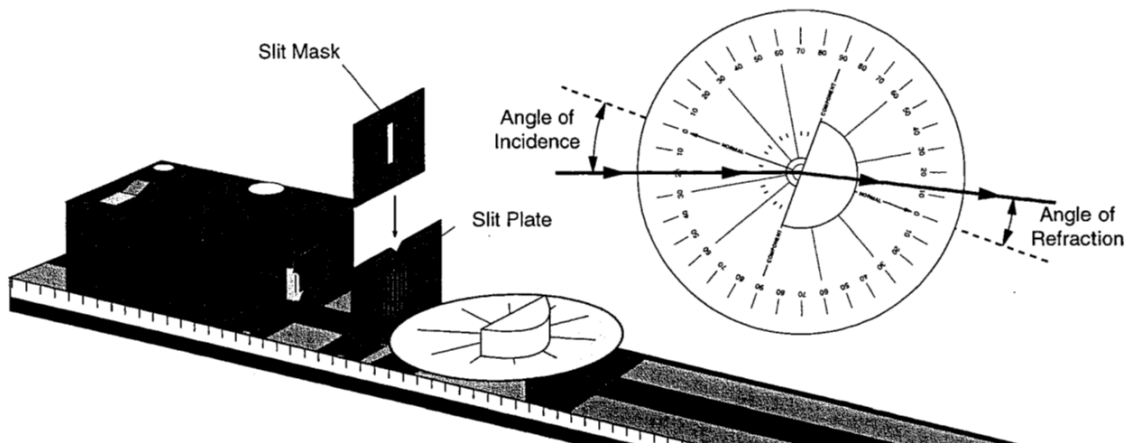


Refraction Lab

This lab is based on Experiment #4 in PASCO's Introductory Optics System Manual and the first illustration is borrowed from it.

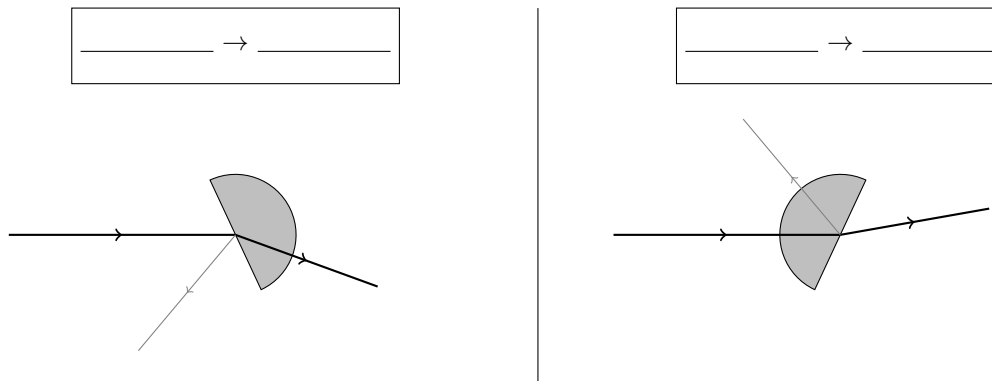
Purpose

To observe what happens to rays of light refract at a lens



Procedure

1. Set up the experiment as indicated above.
2. Adjust the components so that a single ray of light is aligned with the bold arrow labeled “normal” on the Ray Table Degree Scale.
3. Carefully align the flat part of the Cylindrical Lens along the bold arrow labeled “component” on the Ray Table Degree Scale. Adjust the placement of the lens so that the angle of refraction is 0° . From this point forward, be careful when adjusting the Ray Table so that the lens does not slip.
4. Rotate the Ray Table so that the Angle of Incidence changes. Record the angle of refraction and the angle of reflection.
5. We will need to take data for both **Air** $\rightarrow$ **Glass** Refraction and **Glass** $\rightarrow$ **Air** Refraction. Label each situation and label the incident, reflected, and refracted ray.



Name:

Date:

Period:

Data

Not all of the rays will be visible. If a ray is not visible, write “n.v.” for “not visible.”

Angles for Air → Glass		
Incidence	Refraction	Reflection
10°		
20°		
30°		
40°		
50°		
60°		
70°		
80°		

Angles for Glass → Air		
Incidence	Refraction	Reflection
10°		
20°		
30°		
40°		
50°		
60°		
70°		
80°		

Analysis

1. Take a look at how the light bends. It only bends at the flat surface; it does not bend at the curved surface of the lens. Why do you suppose this is?
2. The results of the two trials (**Air** → **Glass** and **Glass** → **Air**) are not the same. What pattern do you notice with the angles of refraction? (*i.e.* Are they larger or smaller than the angle of incidence?)
3. What did you notice about the angle of reFLEction?

Name:

Date:

Period:

4. What difficulties did you have at large angles? Why do you think these difficulties arose?

5. Calculate the index of refraction for the lens at each angle. In theory, these should be the same. You can take an average to find your reported index of refraction. Remember your equation,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 .$$

Be careful when you do **Glass** $\rightarrow$ **Air**. Think carefully about which index is n_1 and which is n_2 .

Air ($n_1 = 1$) $\rightarrow$ Glass ($n_2 = ?$)		
θ_1	Measured θ_2	Measured n_2
10°		
20°		
30°		
40°		
50°		
60°		
70°		
80°		
Average		

Glass ($n_1 = ?$) $\rightarrow$ Air ($n_2 = 1$)		
θ_1	Measured θ_2	Measured n_1
10°		
20°		
30°		
40°		
50°		
60°		
70°		
80°		
Average		

Measured index of refraction (Average of **Air** $\rightarrow$ **Glass** and **Glass** $\rightarrow$ **Air**): _____

Name:

Date:

Period:

6. The accepted value for the index of refraction of the acrylic that the lens is made from is 1.495. Calculate your percent error.

$$\text{Percent Error} = \frac{|\text{measured} - \text{expected}|}{\text{expected}} \times 100\%$$

7. Using your *measured* (not the expected) index of refraction, calculate the critical angle of your acrylic lens. Be careful in determining what n_1 and n_2 are. Remember, total internal refraction only happens **Glass** $\rightarrow$ **Air**; it does not happen **Air** $\rightarrow$ **Glass**.

8. Does your measured critical angle match your data? Explain why it does or does not.